

dashboard and control panel, attackers could configure the IP, port, and duration of the attack. Hackers simply need to type the Website URL they wish to attack, and they can instantly disable targeted sites. Here are some points on why Web servers are used in DDoS 2.0:

- Servers provide a powerful DDoS attack platform, because they usually have greater bandwidth than a simple PC.
- Servers are always online, while a typical PC might go offline. Moreover, they are also rarely formatted.
- A Web server's outgoing traffic is usually less monitored by ISPs, because of a common misconception that a server's outgoing traffic is not as malicious as a PC's.
- By using Web servers as zombies, attackers are even less detectable, because trace-backs typically lead to a lone server at a random hosting company.

## The tool of choice in DDoS 2.0

Do you recall the Wikileaks incident, when attacker groups supporting Wikileaks made DDoS attacks on servers of organisations that opposed Wikileaks? The tool that topped the suspected list of attack weapons was LOIC (Low-Orbit Ion Cannon). In the current era of DDoS 2.0, LOIC is one of the first choices of attackers. It is an open source network-attack application written in C#, which performs DoS/DDoS attacks on a target site by flooding the server with TCP packets, UDP packets, or HTTP requests. An attacker downloads the LOIC client and configures it to connect to an IRC server. The victim server gets flooded with requests from all LOIC clients, operating in 'hive' mode. This is a classic Distributed Denial of Service (DDoS) using a bot-net, except that in this case, attackers volunteer to join it. If you are using this tool even for testing purposes, please be careful, because it does not include code for masking the originator's IP address, which will show up on the target server's logs and can easily be traced back to the user's ISP account, and eventually the local router. But if you want to test it on your own servers, the C# code is available at <https://github.com/NewEraCracker/LOIC/>. In LOIC, most of the files are for creating the interface, but three of them are of interest: *frmMain.cs*, *HTTPFlooder.cs* and *Program.cs*.

The file *frmMain.cs* generates the main part of the user interface, and where the user specifies the URL or IP address of the target server, the program does a series of checks for valid addresses, port numbers, payload, etc, before running the DDoS code for whichever of the three methods (TCP, UDP or HTTP) is selected. In the 'Hive' mode, commands are sent to the LOIC client through IRC. The IRC server, channel and port are set initially in the forms and defined in *Program.cs*, which uses the *C# SmartIRC4NET* library. In LOIC's default mode, the user has volunteered to join the rest of the LOIC users all over the world, thus forming a bot-net, which collectively sends mass requests to the target server.

If you face some difficulty in compiling LOIC, you can go for its binary at <http://sourceforge.net/projects/loic/>. However, besides LOIC, attackers also use a variety of other tools. Some of them are available at <http://packetstormsecurity.org/distributed/>.

## Why DDoS attacks are difficult to detect

If a server is under a DDoS attack, it is quite difficult to detect the attack before the damage is done. The following characteristics of DDoS attacks make them very effective for attackers, and extremely challenging to defend against:

- Attack packets usually have spoofed source IP addresses. Hence, it is more difficult to trace them back to their real source. Further, it is possible that intermediate routers and ISPs may not cooperate in this attempt. Sometimes attackers, by spoofing source IP addresses, create counterfeit armies.
- The similarity of the attack traffic to legitimate traffic makes separation and filtering extremely hard. Unlike other security threats that need specially crafted packets (e.g., intrusions, worms, viruses), flooding attacks need only a high traffic volume, and can vary packet contents and header values at will.
- The easy availability of powerful DDoS tools has made it possible for even inexperienced users to bring down mighty Web servers.



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